

CV Project: Pedestrian Safety at SFU Report

IAT 360

Group 4

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Data Links:

All Data Combined:

 CV Project ALL DATA

Emily's Collected Data:

 Emily's Data

Zoe's Collected Data:

 Zoe's Data

Introduction

This project is focused on crosswalk safety at Simon Fraser University's Burnaby campus. The problem we are trying to solve is reducing risk for pedestrians trying to cross roads at marked crosswalks. This is a detection problem using supervised learning. We are using this model to detect hazards at crosswalks, including "don't walk" signs, vehicles, and cyclists. We are also detecting crosswalks, "walk" signs, and traffic signals.

Use Cases: The use case of our project would be for anyone trying to navigate SFU Burnaby safely on foot. Specifically, this project is meant to improve safety at SFU crosswalks. This particular model could also be very useful for the visually impaired.

Target Audience: The target audience of our project would be students, staff, or anyone else travelling on foot at the SFU Burnaby campus. This model can also be useful for the visually impaired or those with mobility issues.

Devices Where the Model Can Be Used: This model could potentially be used on a few devices. The model could work on a smartphone using the camera feature or maybe even a GoPro. The model could also work on wearable technology like smart glasses that feature cameras.

Dataset Report

How We Collected and Created Our Data:

Our external data was collected from three separate datasets to ensure we had an adequate number of samples for each class:

Cyclist Set: <https://universe.roboflow.com/cycler-vi9vn/cyclists-lt9pl>

Crosswalk Sign Set: <https://universe.roboflow.com/walkassist/walk-assist/browse?queryText=&pageSize=50&startingIndex=100&browseQuery=true>

Vehicle Set: <https://universe.roboflow.com/faculty-of-engineering-alexandria-university-mvdnh/vehicles-sgibq>

The data we collected ourselves was taken with mobile phone cameras on two separate days at the SFU Burnaby campus. We took images on two separate days, hoping to capture different weather conditions. Images were taken at crosswalks of various vehicles, cyclists, crosswalk markings, traffic signals, and walk/don't walk signs. In total, we took 89 images around the SFU Burnaby Campus. We then labelled all our data using Label Studio. We labelled the data with the same categories that were featured in the datasets we collected from online.

We did have to remove one category from the external dataset that was called “threewheel”. This category featured a vehicle that we had never seen in Canada, so it was not appropriate for our dataset. We also removed any images that were uncategorized in the external dataset. After all our data was collected and combined, we used Roboflow to create an 80/20 split for the whole dataset. In total, we have 3910 images in our dataset.

Annotation Tool We Used

Label Studio: <https://labelstud.io/>

Our Labels:

- Crosswalk
- Don't Walk
- Traffic Signal
- Walk
- Bikerider
- Bus
- Car
- Motorbike
- Truck
- Van

Example Images of New Data We Added: **Crosswalk:**



Car + Crosswalk:



Don't Walk + Traffic Signal:



Bus + Crosswalk:



Bikerider + Crosswalk:



Walk Sign + Traffic Signal:



Dataset Biases and Privacy:

In the data we collected, there is representation bias. Hazards, including bike riders, buses, cars, motorbikes, trucks, and vans, are overrepresented due to their large, diverse datasets. However, the dataset, including crosswalks, don't walk and walk signs, and traffic signals, contained significantly fewer images. These classes were underrepresented in our data. This bias could have been fixed by balancing the number of images per class, adding more samples to our signage and crosswalks dataset. However, we could not find much online data containing signage or scenery that matched the SFU Burnaby campus. There was only one dataset we could find with similar signs that appeared to be taken in North America.

A major concern while collecting our own data was capturing and identifying information of people in our images, such as faces and license plates. As the images are not being used for commercial purposes, people in public spaces can be photographed.

Another privacy concern was ownership rights for the online datasets we used. As these datasets were all public and open source, there are no issues regarding our usage of them.

Training Report

Initial Training

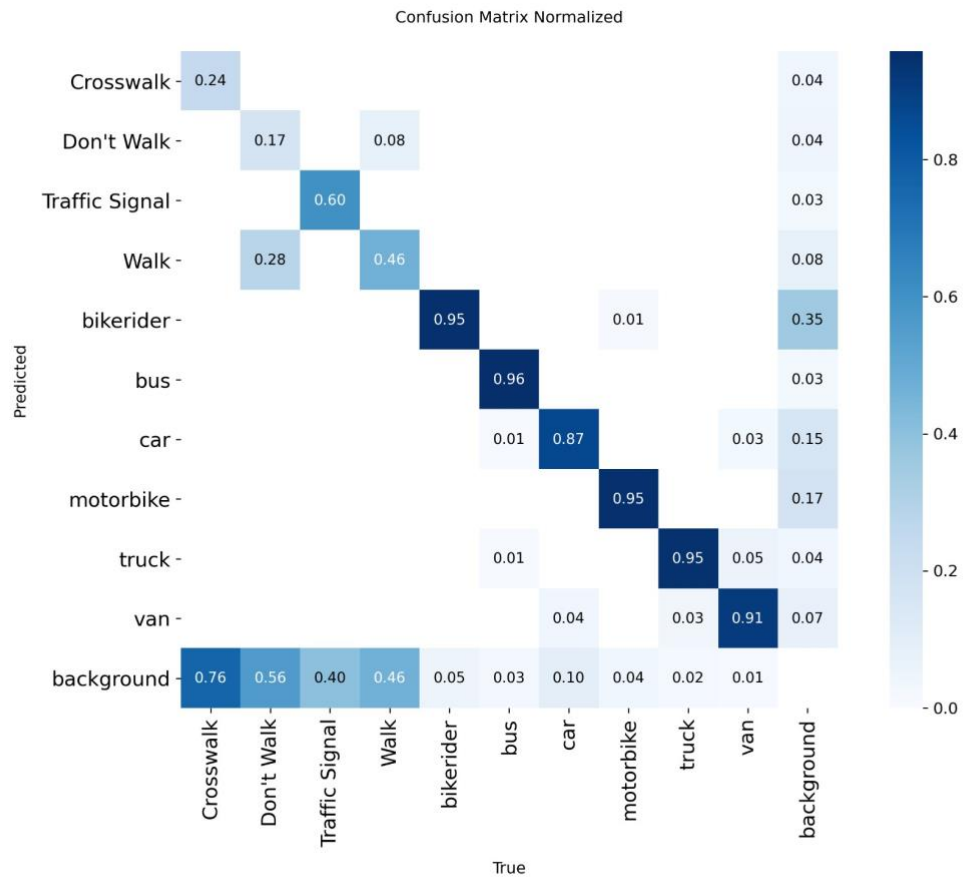


Figure 1: Confusion matrix after initial training and evaluation

```

Ultralytics 8.3.221 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 11,129,454 parameters, 0 gradients, 28.5 GFLOPs
val: Fast image access (ping: 0.6±0.2 ms, read: 22.1±14.6 MB/s, size: 38.7 KB)
val: Scanning /content/drive/MyDrive/CV Project Oct 24/labels/val.cache... 788 images, 0 backgrounds, 0 corrupt: 100%

```

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95)
all	788	1080	0.76	0.684	0.721	0.584
Crosswalk	22	25	0.602	0.16	0.341	0.168
Don't Walk	16	18	0.645	0.204	0.251	0.08
Traffic Signal	11	15	0.603	0.507	0.47	0.291
Walk	13	13	0.39	0.385	0.425	0.293
bikerider	251	414	0.897	0.935	0.962	0.709
bus	103	119	0.983	0.958	0.977	0.918
car	125	134	0.849	0.858	0.903	0.871
motorbike	104	137	0.824	0.92	0.932	0.712
truck	75	92	0.917	0.96	0.978	0.867
van	103	113	0.894	0.956	0.974	0.935

```

Speed: 1.1ms preprocess, 7.7ms inference, 0.0ms loss, 1.6ms postprocess per image
Results saved to /content/runs/detect/val
np.float64(0.6841793578489189)

```

Figure 2: Metrics after initial training and evaluation

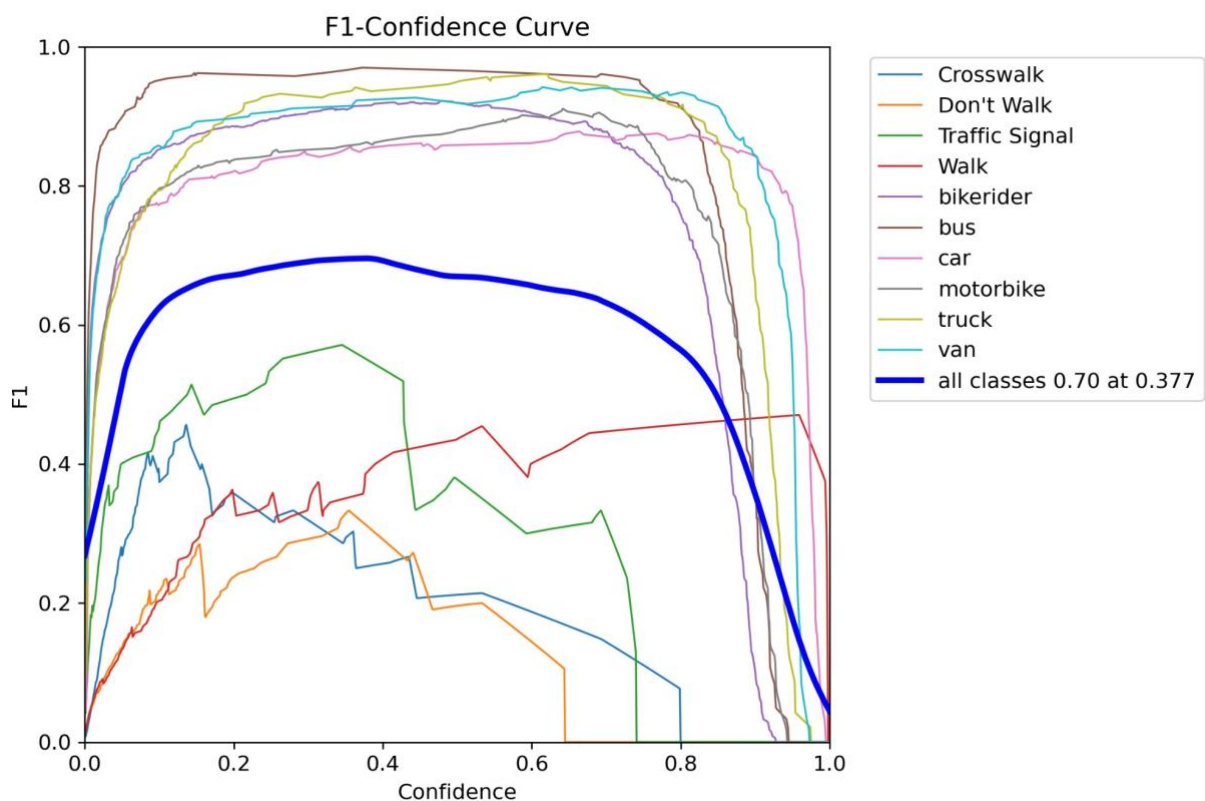


Figure 3: F1-confidence curve after initial training and evaluation

From the outcome of the initial training and evaluation, we can determine that the model was accurate in detecting hazards: cyclists, buses, cars, motorbikes, trucks, and vans. This is a good sign, as these hazards threaten pedestrian safety, and we want the model to be able to predict them accurately.

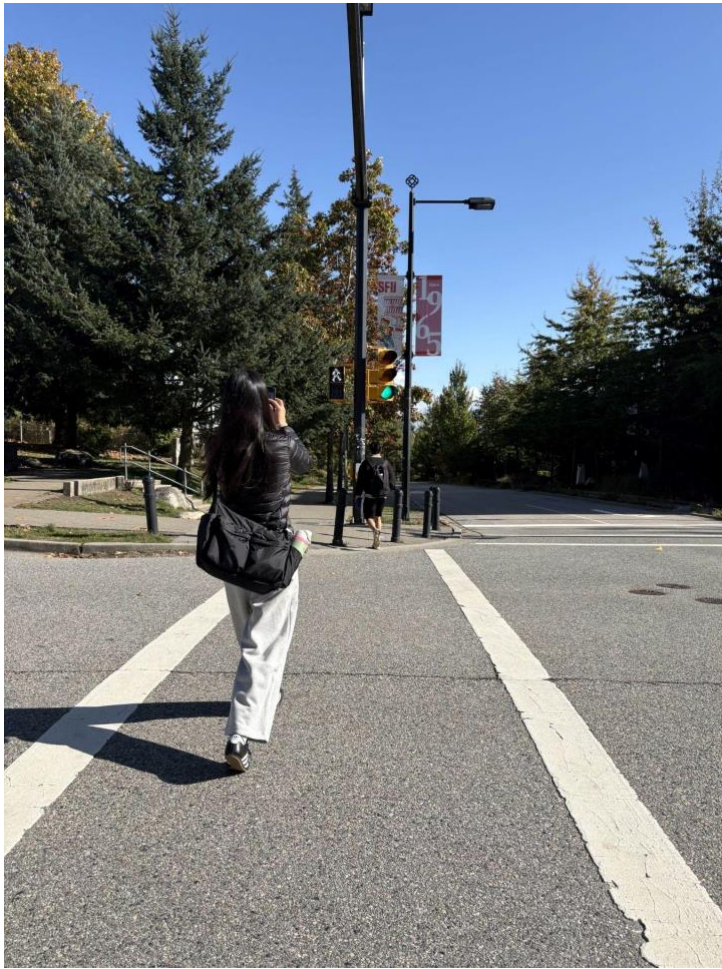
The model, however, did not perform as strongly for crosswalks, Don't Walk signs, traffic signals, and Walk signs. We concluded that our model was likely underfit in these categories. We tried to fix this by increasing the Epoch to 15 and Patience to 15.

Example Images From Original Labeling and Initial Run:

This image only worked on the car:



Nothing was detected in this image:



From these images, you can see that our model was having issues detecting most categories. But it worked well with cars. We concluded this was likely because our vehicle dataset was bigger than the other categories' datasets. We also think crosswalks, in particular, are not being detected because they fall more under a segmentation problem category. Crosswalks that are not Zebra crosswalks in particular likely cause problems due to their similarity to background elements.

These images below were from our original labeling but with Epoch 15 and Patience 15:



As seen in the images above, there was an improvement with detecting "Walk" and "Don't Walk" signs. However, there was still some incorrect labeling occurring with the model.

Changing Epoch to 15 + Patience to 15 + Fixing Labeling in Our Dataset:

These results show what happened to our model after increasing the epoch and patience to 15. We chose 15 because we had good results after increasing from 5 to 10, but worse results when we increased to 20. We found 15 to be the best spot. We also decided to fix the labels in the dataset. Some of the data we collected from the Burnaby campus was missing a few labels, so we added those. We also fixed our “Walk” and “Don’t Walk” sign labels to include the bottom countdown timer. We previously had issues with the timer being labelled incorrectly as a separate “Walk” or “Don’t Walk” sign.

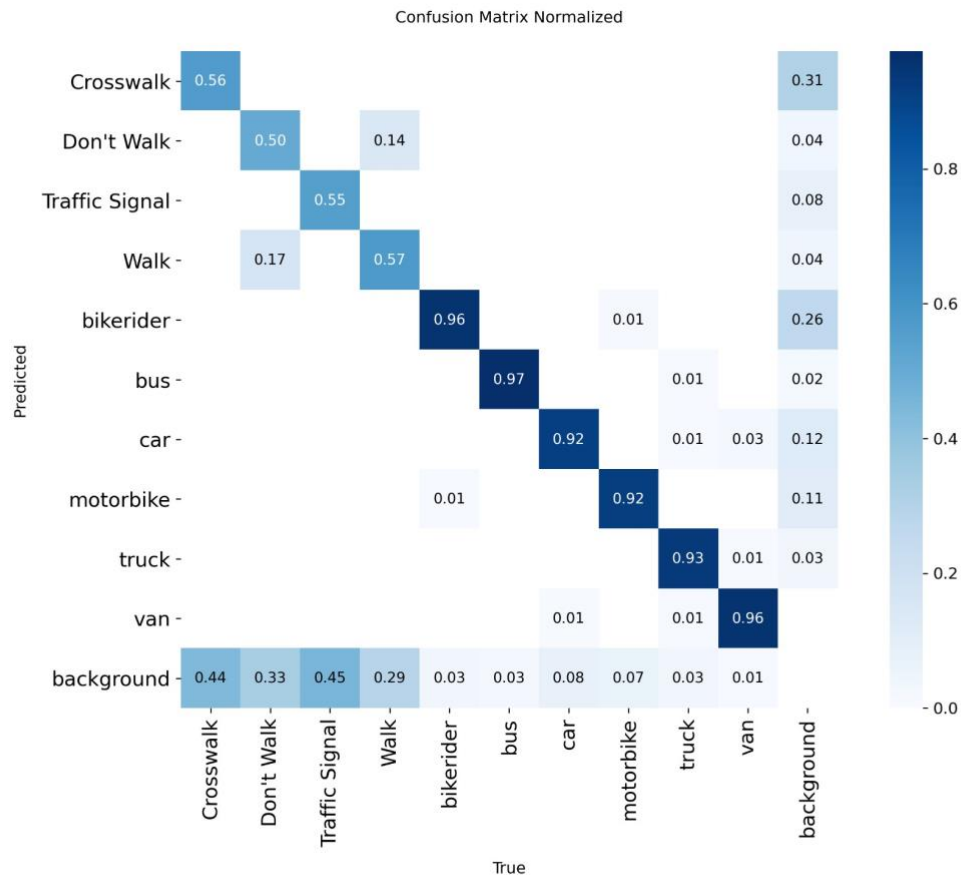


Figure 4: Confusion matrix after changing Epoch to 15 and Patience to 15

```

Ultralytics 8.3.223 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 11,129,454 parameters, 0 gradients, 28.5 GFLOPs
val: Fast image access (ping: 0.0±0.0 ms, read: 1525.7±801.0 MB/s, size: 67.7 KB)
val: Scanning /content/CV-Project-Nov-1-2/labels/val.cache... 788 images, 0 backgrounds, 0 corrupt: 100% 788/788 1.5Mit/s 0.0s
50/50 4.2it/s 11.8s

```

Class	Images	Instances	Box(P)	R	mAP50	mAP50-95
all	788	1090	0.855	0.736	0.793	0.636
Crosswalk	30	32	0.632	0.312	0.429	0.18
Don't Walk	16	18	0.897	0.5	0.606	0.318
Traffic Signal	14	20	0.588	0.4	0.377	0.211
Walk	14	14	0.728	0.574	0.709	0.42
bikerider	250	412	0.944	0.938	0.969	0.77
bus	103	119	0.98	0.95	0.992	0.944
car	124	133	0.907	0.887	0.931	0.896
motorbike	104	137	0.922	0.912	0.955	0.766
truck	75	92	0.974	0.935	0.975	0.903
van	103	113	0.981	0.956	0.986	0.956

Figure 5: Metrics after changing Epoch to 15 and Patience to 15

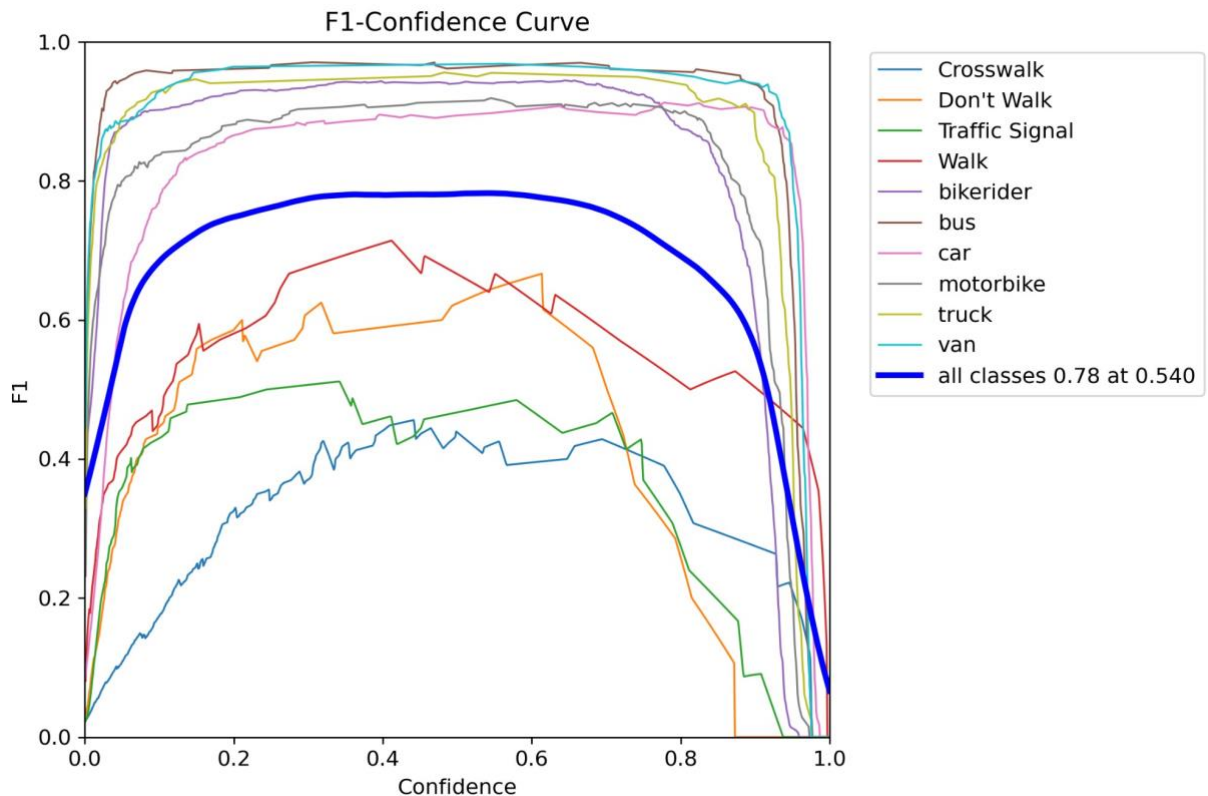


Figure 6: F1-confidence curve after changing Epoch to 15 and Patience to 15

The results from this run showed great improvement from the original run. The “Walk”, “Don’t Walk” and “Crosswalk” categories greatly increased in confidence score. Except the “Traffic Signal” category slightly decreased in improvement.

Hyperparameter Change:

For our hyperparameter change we decided to change the learning rate from 0.0005 to 0.005 and imgsiz from 640 to 500.

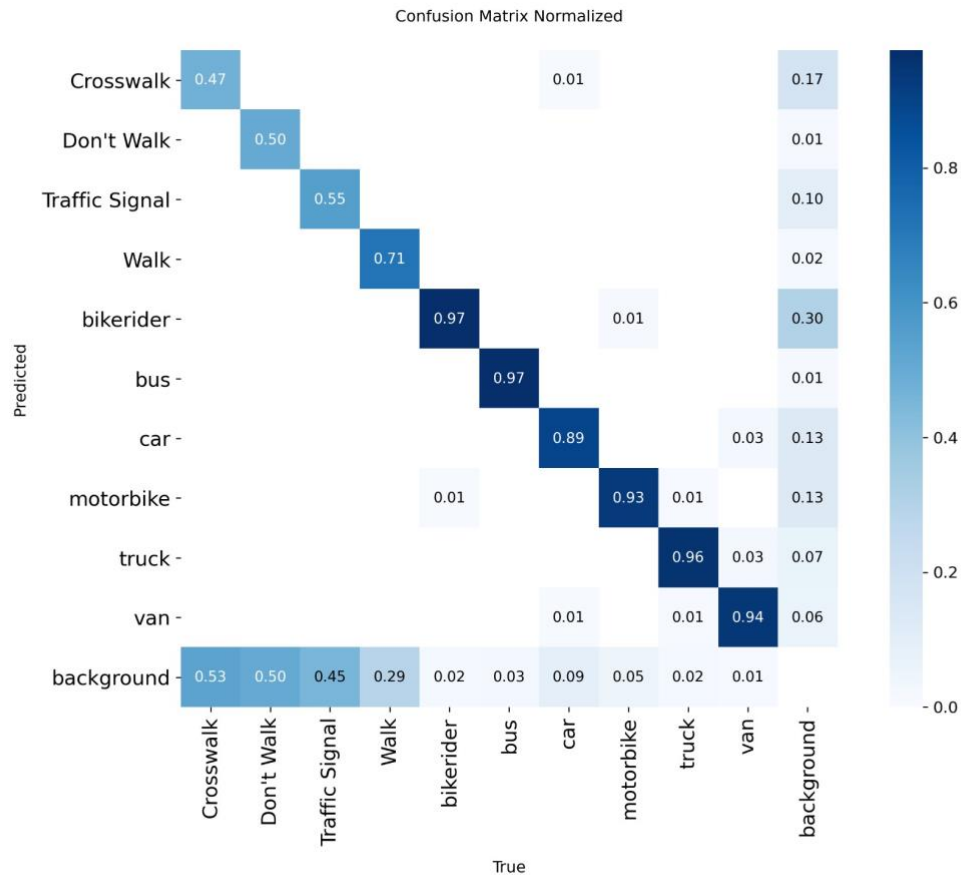


Figure 7: Confusion matrix after changing Learning Rate to 0.005 and Imgsz to 500

```
Ultralytics 8.3.223 Python-3.12.12 torch-2.8.0+cu126 CUDA:0 (Tesla T4, 15095MiB)
Model summary (fused): 72 layers, 11,129,454 parameters, 0 gradients, 28.5 GFLOPs
val: Fast image access (ping: 0.0±0.0 ms, read: 1212.7±958.1 MB/s, size: 67.7 KB)
val: Scanning /content/CV-Project-Nov-1-2/labels/val.cache... 788 images, 0 backgrounds, 0 corrupt: 100% 788/788 333.3Kit/s 0.0s
Class      Images  Instances  Box(P  R      mAP50  mAP50-95): 100% 50/50 4.9it/s 10.1s
all         788      1090      0.853  0.781  0.789  0.629
Crosswalk   30       32       0.627  0.368  0.41  0.151
Don't Walk  16       18       0.499  0.499  0.557  0.256
Traffic Signal 14       20       0.58  0.55  0.428  0.227
Walk        14       14       0.781  0.714  0.702  0.443
bikerider   250      412      0.921  0.968  0.969  0.769
bus         103      119      0.991  0.975  0.993  0.938
car         124      133      0.883  0.88  0.918  0.891
motorbike   104      137      0.907  0.926  0.963  0.77
truck       75       92       0.918  0.978  0.967  0.884
van         103      113      0.927  0.947  0.984  0.966
```

Figure 8: Metrics after changing Learning Rate to 0.005 and Imgsz to 500

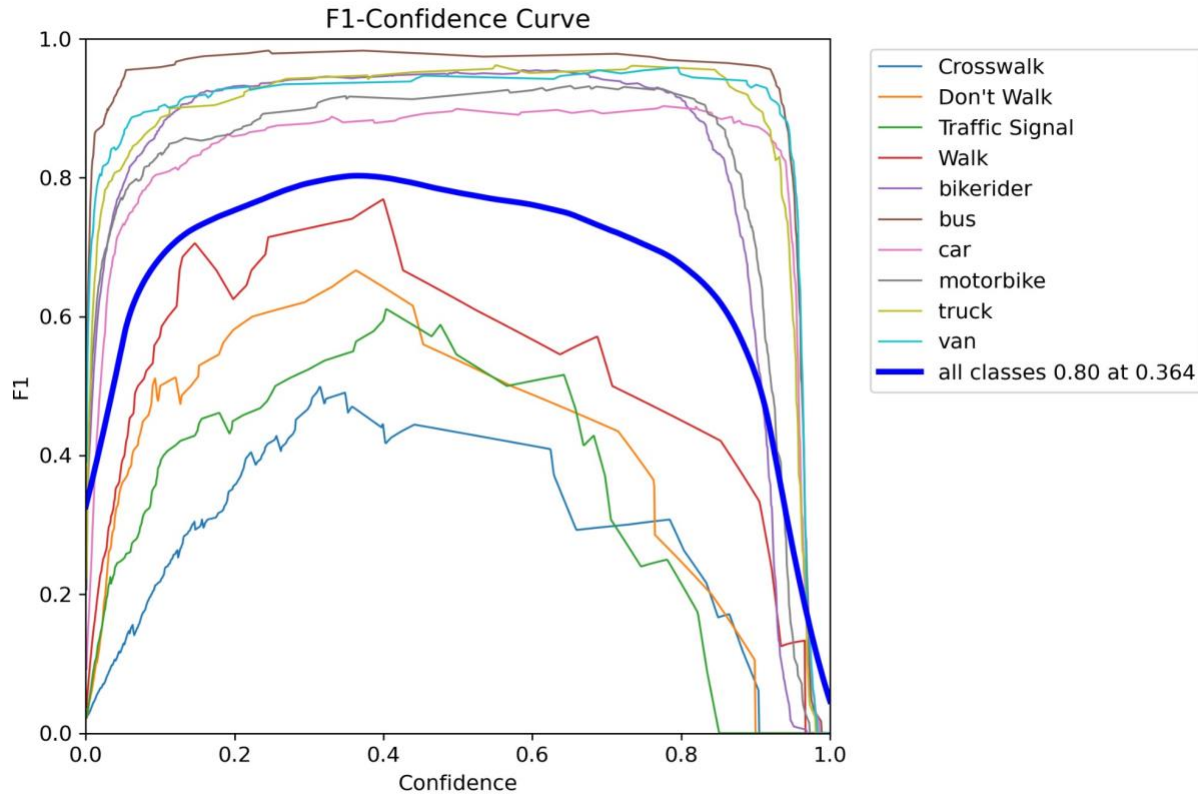


Figure 9: F1-confidence curve after changing Learning Rate to 0.005 and Imgsz to 500

We decided to increase the learning rate from 0.0005 to 0.005. We wanted to experiment with a faster learning rate to see if the model would learn more at each step. With a higher learning rate, we risk overshooting, but could improve learning efficiency. The learning rate determines how much the model adjusts its parameters at each step while training.

We reduced the image size from 640 to 500. The image size determines the size images are resized to during training, so the model sees each image at a consistent size. Therefore, our model would be seeing 500x500 images. This can have an impact on model accuracy, and we wanted to see the effect of making images smaller. Typically, larger images would allow for recognition of farther objects and more accuracy, but the tradeoff is slower training and higher memory usage. With smaller image sizes, there are fewer pixels to process, giving us faster training and less memory usage. We were curious if, for our case, simpler images with fewer pixels would help remove any visual complexity. The backgrounds of our images were complex, so we wanted to see if this change could potentially help the model identify our chosen objects better.

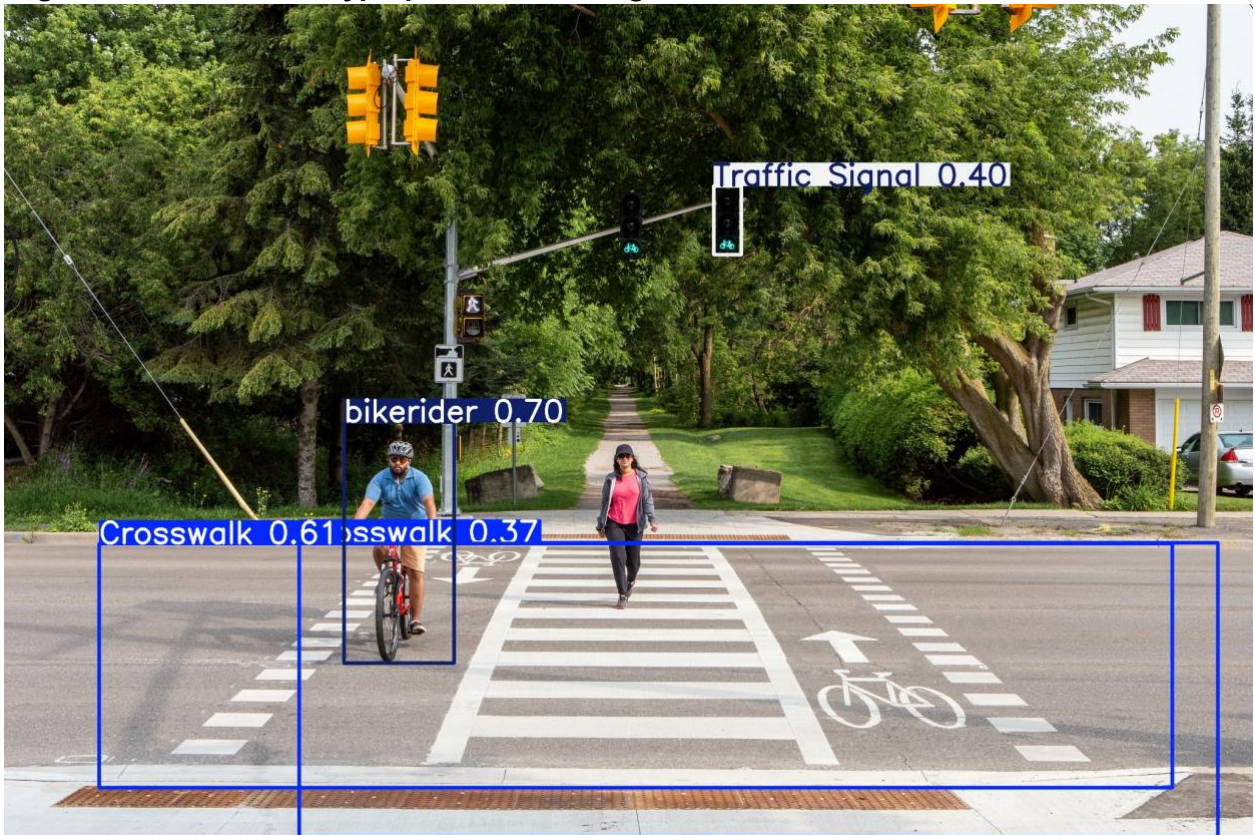
As seen in the confusion matrix, there was a significant improvement with "Walk" signal detection. However, there is a slight decrease in crosswalk detection. We believe this trade-off

was okay because “Walk” signal detection is more important to safety. If the model detects a “Walk” signal, that means you are very likely already at a crosswalk. Previously, “Walk” and “Don’t Walk” were getting confused with each other, which, according to our confusion matrix, is no longer an issue after our hyperparameter change.

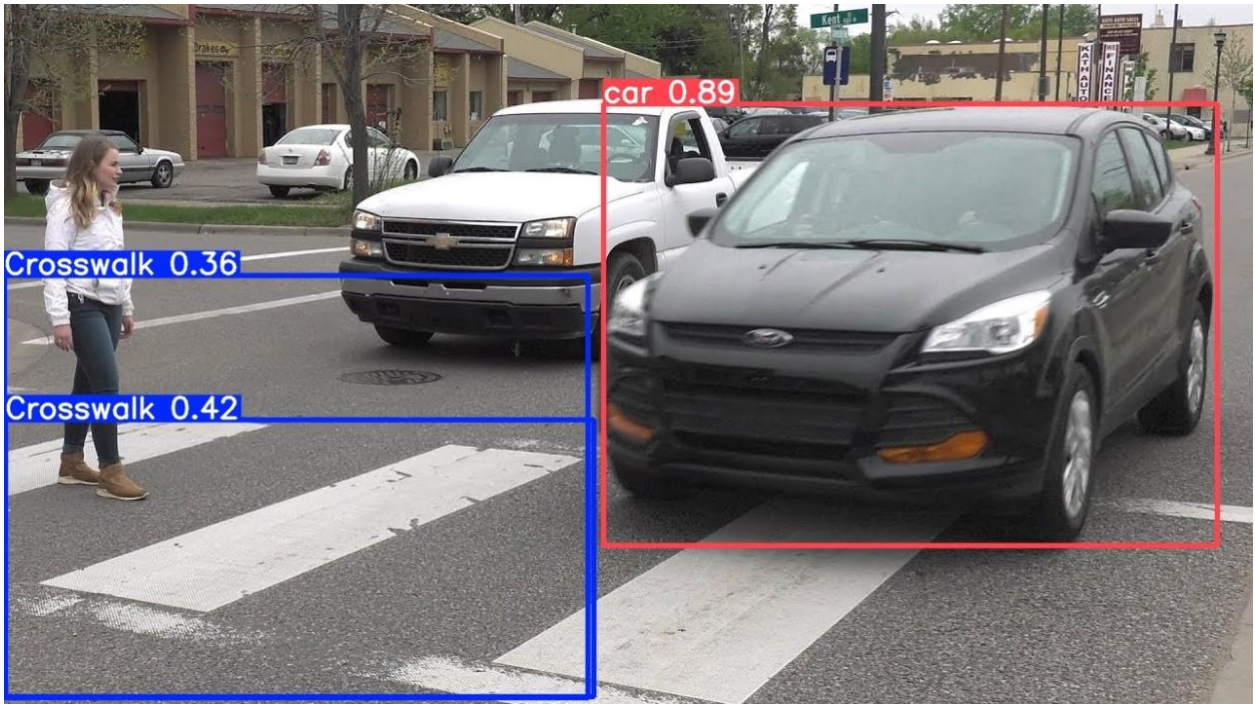
We noticed a difference in training time after changing hyperparameters. The model trained much faster. We predict that the model performed slightly better overall due to the reduction in image size. Our images had complex backgrounds because they were taken outside and had many objects within them. Reducing image size may have simplified the images and allowed the model to better focus on the objects we wanted to identify.

Overall, we believe our model was tuned to the best of our abilities and are very proud of the results. The images below show how well our model works.

Image Predictions After Hyperparameter Change:



The model managed to predict the crosswalk, the cyclist, and even the traffic signal despite it being far away in the image. It did not detect the walk sign, likely because it is too far away.



In this image, the model detected the crosswalk and one of the vehicles. However, it failed to detect the truck on the left. We are wondering if this is because the black car is blocking the truck partially.



Here, the model detected the crosswalk. However, it failed to detect the car. We think this is because the datasets used mainly consisted of black, white, or grey cars. It did not contain many colourful or blue cars. It also detected a pedestrian walk sign even though this is different from the walk signals. This could cause future issues in our model.



In this image, the model detected the crosswalk and traffic signal. It did not detect the truck. This could be because the truck is too far away, and the model has identified it as background.



Here, the model has successfully detected the don't walk sign and the traffic signal. This is a clear, close up image, making it easier for the model to detect.



In this image the model has successfully detected the walk sign with high confidence. This is because it is a clear, close up image of a walk sign.

As seen in the images above, our model works quite well. We believe after changing our hyperparameters we have improved it to the best of our abilities. However, the model still has flaws. Seen below is an example of one of our model's flaws.



Our model still seems confused with the above image we had tested previously. However, it is slightly less confident in the Walk prediction which is good.



But it does really well with this image. Overall, it appears there is still an issue with blank signal boxes. We believe this could be because we were only able to relabel our own data, which we collected on the Burnaby campus and not the external data. The external data was too large to manually relabel. However, if this model were ever to be actually used, we would need to

ensure this problem is fixed. This issue could create danger for the user, as there are conflicting messages on whether or not it is safe to walk.

Challenges We Faced:

We used Roboflow to combine our three datasets, as well as our own collected data. However, due to some external datasets' default settings, a couple of datasets we used had data only categorized as training. This meant that when we combined datasets, we had classes that were imbalanced between the train and validation sets. For instance, only the training set contained images with crosswalk, don't walk/walk sign, and traffic signal labels. None of these classes appeared in the validation set. To solve this problem, we fixed each dataset split separately in Roboflow by manually splitting them into a train and validation set. We then combined all sets again and made sure the resulting dataset had an 80/20 split too.

Another challenge when collecting data was waiting for cars and cyclists to pass by the crossing areas. It was not busy, and so we had a harder time finding hazards to photograph. We also took pictures on two separate days to overcome the challenge of gathering data with only one weather condition.

It was also difficult for us to find external data sources that matched how the SFU Burnaby campus looked, as well as its signs. Therefore, there was representation bias among the classes, which made the model have a bias towards certain hazards. The model identified vehicle hazards more accurately than crosswalks and signs. This is because we only found one smaller dataset containing signs and scenery that matched the SFU Burnaby campus.

Another challenge was that the shape of crosswalks made it difficult to draw bounding boxes around them. The crosswalk bounding boxes often included the crosswalk markings, but also large pieces of background. This explains why they are often confused with the background.



The biggest issue we faced was that our model sometimes classified instances of “walk” and “don’t walk” signs in the same image. This can confuse pedestrians and potentially put them in danger. This could be due to the walk and don’t walk signs being similar shapes, as well as in this image, the faint outline of the numbers could be mistaken for the white lights of the walk sign. However, the model is much more confident in its don’t walk prediction than it is with its walk sign prediction.

To try reducing this issue, we relabeled the walk and don’t walk signs in our data so that the entire sign was within the bounding box:



Before



After

After making this correction, although the confusion matrix showed the same accuracy for don't walk sign detection, the accuracy for walk detection increased significantly.

References

Cyclist Set:

<https://universe.roboflow.com/cycler-vi9vn/cyclists-lt9pl>

Crosswalk Sign Set: <https://universe.roboflow.com/walkassist/walk-assist/browse?queryText=&pageSize=50&startingIndex=100&browseQuery=true>

Vehicle Set:

<https://universe.roboflow.com/faculty-of-engineering-alexandria-university-mvdnh/vehicles-sgibq>

Annotation Tool (Label Studio):

<https://labelstud.io/>